



DAM

WEB INTERFACE DESIGN

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Profesora:

Begoña Rodríguez Ferreras

begona.rodfer.1@educa.jcyl.es

BOOK



Diseño de interfaces web

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USER INTERFACE DESIGN FOR WEB DEVELOPMENT

Goals:

- Recognize the importance of visual communication and its basic principles.
- Analyze different alternatives for presenting information in web documents.
- Identify and analyze the elements involved in creating web prototypes.
- Use and analyze content management systems as the foundation for developing a website.
- Plan and establish the process for designing a website.

DESIGN PRINCIPLES BASED ON HUMAN-COMPUTER INTERACTION

Human-Computer Interaction (HCI) is a field of study that focuses on the design and development of interactive systems that are easy and efficient to use.

HCI guidelines are recommendations that can be used to design and develop user interfaces that are more user-friendly and effective.

LICKLIDER & CLARK (1962)

Ten **problems to solve** for better Human-Computer Interaction (HCI):

- Sharing computer time among many users.
- An input-output system for communication using symbolic and graphical data.
- An interactive system capable of processing operations in real time.
- Systems for massive information storage that allow rapid retrieval.
- Systems that facilitate cooperation among people in the design and programming of large systems.

LICKLIDER & CLARK (1962) (II)

Ten **problems to solve** for better Human-Computer Interaction (HCI) (II):

- Recognition by computers of voice, printed handwriting, and direct handwritten data input.
- Automatic translation between natural languages.
- Automatic programming — enabling computers to generate or assist in writing programs.
- Self-organizing systems capable of adapting and improving their own performance.
- Logical and heuristic problem-solving systems that support human reasoning.

LICKLIDER & CLARK (1962) (III)

These ten problems anticipated almost everything modern computing tries to solve:

- multitasking
- graphical interfaces
- real-time systems
- databases
- collaborative development
- speech recognition
- machine translation
- AI-assisted coding
- adaptive systems
- expert systems

HANSEN (1972)

Wilfred J. Hansen's four principles for user-interface design, proposed in 1971, they are an important early contribution to Human-Computer Interaction (HCI).

Hansen presented them in “User Engineering Principles for Interactive Systems”, published in the proceedings of the 1971 Fall Joint Computer Conference.

The four principles are:

1. Know the user
2. Minimize memorization
3. Optimize operations
4. Engineer for errors

1. KNOW THE USER

The interface should be designed around the people who will actually use it, rather than around what is technically convenient for the programmer.

The designer should understand:

- the user's goals and needs;
- their previous experience;
- their knowledge and skills;
- the context in which they work;
- their limitations and likely difficulties.

2. MINIMIZE MEMORIZATION

Users should not have to remember unnecessary information in order to operate the system.

The interface should provide information that the user needs at the moment they need it.

For example, instead of requiring:

"Remember that Ctrl+Alt+N creates a new network."

the system could provide a visible New network button or menu item.

This principle is closely related to a major modern HCI idea: recognition is generally easier than recall.

2. MINIMIZE MEMORIZATION (II)

You can see its influence in things we now take for granted:

- menus;
- visible buttons;
- autocomplete;
- contextual help;
- suggestions;
- recently used items;
- meaningful icons and labels.

3. OPTIMIZE OPERATIONS

The system should make users' actual tasks efficient.

This means reducing unnecessary actions, avoiding cumbersome procedures and designing operations around the user's workflow.

Design the system so that the user's work can be accomplished efficiently and naturally.

This is particularly relevant today for usability, keyboard shortcuts, automation, defaults and reducing the number of steps required to accomplish frequent tasks.

4. ENGINEER FOR ERRORS

Hansen recognized that users will make mistakes, and that a good system should therefore be designed to anticipate and handle them.

The important point is that errors should not simply be treated as the user's fault. For example, if a user enters: 192.168.1.300
a good interface should detect the invalid address and explain the problem.

An even better interface might prevent invalid input from being entered in the first place.

4. ENGINEER FOR ERRORS (II)

This principle leads to several modern practices:

- validation.
- meaningful error messages.
- confirmation for dangerous operations.
- undo.
- recovery mechanisms.
- preventing invalid actions.
- preserving user data after an error.

CORE FUNCTIONS OF GRAPHIC DESIGN

Graphic design isn't just "making things pretty." In web development, it has functional, communicative, and cognitive purposes.

- **Communication & Information Delivery:** Translates raw data, copy, or abstract concepts into structured, easily digestible visual messages (think infographics, road signs, or annual reports).
- **Persuasion & Marketing:** Drives action and influences audience behavior in advertising, campaigns, and product packaging by appealing to specific emotions and needs.
- **Identity & Brand Building:** Establishes the distinct visual personality of a company, product, or organization through consistent logos, typography, color schemes, and imagery.

CORE FUNCTIONS OF GRAPHIC DESIGN (II)

Graphic design isn't just "making things pretty." In web development, it has functional, communicative, and cognitive purposes.

- **User Experience & Navigation:** Guides people through digital spaces (websites, apps) and physical environments (wayfinding in airports or subways) by creating intuitive visual hierarchies.
- **Engagement & Aesthetics:** Captures attention in a visual-heavy world, making content memorable and pleasant to interact with.

GROUPS OF ELEMENTS

- **Conceptual elements:** The ideas, concepts and meanings that the interface represents. Example: the concept of "deleting a file."
- **Visual elements:** What the user actually perceives: colours, typography, icons, shapes, images, etc. Example: a trash-bin icon representing deletion.
- **Relational elements:** The relationships between elements: hierarchy, proximity, alignment, grouping, sequence and navigation. Example: placing "Save" next to "Cancel", or grouping network settings together.
- **Practical elements:** The elements that allow the user to perform actions and accomplish tasks. Example: buttons, forms, sliders, menus and controls.

DEFINITION OF WEB INTERFACE

For all the above, we can conclude that:

*“A web interface is the set of **visual and interactive elements**, together with their **organization and layout**, that allows users to access information, navigate a website and interact with its content and functions efficiently.”*

If these elements are not carefully designed and coordinated, a website may still work technically, but achieving a good user experience will largely be a matter of chance.

ADDITIONAL READINGS

Guidelines in Human Computer Interface (HCI)

Interaction in HCI: Principles, types, and examples

¿ANY QUESTION?